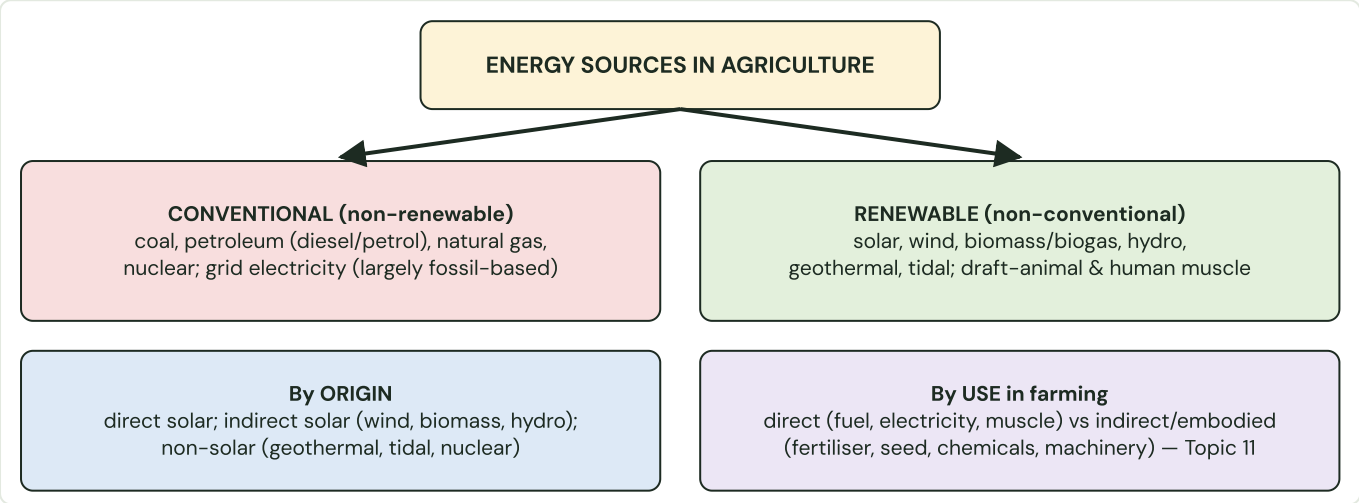
UNIT 7 · TOPIC 1

Conventional and Renewable Energy Sources in Agriculture

Syllabus topic: Conventional and renewable energy sources in agriculture

Classification of energy sources



Classification of energy sources (redrawn from the source figure)

Basis	Classes
RENEWABILITY	CONVENTIONAL (non-renewable, finite): coal, petroleum (diesel/petrol — the tractor fuel), natural gas, nuclear, fossil-based grid electricity. RENEWABLE (non-conventional, replenished): solar, wind, biomass/biogas, small hydro, geothermal, tidal, and human and draft-animal muscle power
ORIGIN	Direct solar (solar thermal/PV); INDIRECT solar (wind, biomass and hydro all trace back to the sun); non-solar (geothermal, tidal, nuclear)
USE in farming (Topic 11)	DIRECT — diesel, electricity, human/animal muscle used on the farm; INDIRECT/embodied — energy used to make fertilisers (the largest indirect input, especially nitrogen), seeds, agro-chemicals and machinery

Agriculture is both an energy CONSUMER and, through biomass, an energy PRODUCER.

Why renewables matter for Indian agriculture

- Diesel and grid power are costly and carbon-intensive.
- Sunshine is abundant: most of India receives **5–7 kWh/m²/day** over ~**300** clear days.
- Solar pumps (PM-KUSUM), biogas and biomass gasifiers cut fuel bills and emissions and improve farm energy security.

Updated information — India renewable energy and solar in agriculture

Item	Status
Installed RENEWABLE capacity (30 June 2026)	≈ **288.58 GW** — third-largest in the world
Mix	SOLAR **162.15 GW** (largest), WIND **57.44 GW**, HYDRO **57.24 GW** (large hydro 52.06 GW + small hydro), the rest bioenergy
Target	**500 GW** non-fossil capacity by 2030 (MNRE/PIB)
Record additions	FY2025-26: ~**44.6 GW** solar (vs 34 GW target) and ~6 GW wind; H1-2026 alone ~29 GW solar + wind; rooftop solar up ~**104%** under PM SURYA GHAR (free rooftop scheme)
PM-KUSUM (cumulative to 30 Nov 2025)	Component-A: **667 MW** decentralised solar plants; Component-B: >**9.42 lakh** STANDALONE solar agricultural pumps; Component-C: >**10.99 lakh** grid-connected pumps SOLARISED
Distributed solar	KUSUM + rooftop ≈ **16.3 GW (36%)** of FY26 additions

The 288.58 GW, 162.15 GW (solar) and 57.44 GW (wind) figures match MNRE data to 30 June 2026 (government reply to Parliament, 22 July 2026). The source listed hydro and "small hydro" separately as well, which double-counts small hydro; the table above gives the breakdown.

ARS/NET links current schemes to fundamentals: remember PM-KUSUM's three components (A — plants, B — standalone pumps, C — solarisation), PM Surya Ghar (rooftop) and the 500 GW-by-2030 target. Recheck figures close to the exam.

Practice MCQs

Q1. Which is a CONVENTIONAL (non-renewable) energy source?

(a) Solar
(b) Wind
(c) Petroleum/diesel
(d) Biogas

Q2. Which is a RENEWABLE energy source?

(a) Coal
(b) Natural gas
(c) Nuclear
(d) Solar

Q3. Wind, biomass and hydro energy are ultimately forms of:

(a) Nuclear energy
(b) Indirect solar energy
(c) Tidal energy
(d) Geothermal energy

Q4. The largest INDIRECT (embodied) energy input in crop production is usually:

(a) Seed
(b) Nitrogen fertiliser
(c) Machinery
(d) Pesticide

Q30. India's total installed renewable-energy capacity (June 2026) is about:

(a) 100 GW
(b) 175 GW
(c) 289 GW
(d) 500 GW

Q31. Solar power is India's largest renewable source at about:

(a) 60 GW
(b) 100 GW
(c) 162 GW
(d) 250 GW

Q32. The scheme promoting solar pumps and solarisation for farmers is:

(a) PM-KUSUM (b) FAME

(c) UJALA (d) SATAT

Q33. PM-KUSUM Component-B provides:

(a) Grid solar plants (b) Standalone solar agricultural pumps

(c) Rooftop solar (d) EV subsidy

Answers

Q1 — (c) Petroleum/diesel

Q2 — (d) Solar

Q3 — (b) Indirect solar energy

Q4 — (b) Nitrogen fertiliser

Q30 — (c) 289 GW — *MNRE data to 30 June 2026: RE 288.58 GW.*

Q31 — (c) 162 GW — *Solar 162.15 GW (30 June 2026).*

Q32 — (a) PM-KUSUM

Q33 — (b) Standalone solar agricultural pumps — *PM-KUSUM B = standalone solar pumps.*

Sources: Sukhatme & Nayak — Solar Energy: Principles of Thermal Collection and Storage; Rai — Non-Conventional Energy Sources; Duffie & Beckman — Solar Engineering of Thermal Processes; WMO Guide to Meteorological Instruments (radiation); MNRE/PIB releases (capacity figures confirmed against MNRE data to 30 June 2026); NPTEL renewable-energy lectures.